ALKENE

1. Addition of bromine to 1, 3-butadiene gives
   (a) 1, 2 addition product only
   (b) 1, 4 addition product only
   (c) Both 1, 2 and 1, 4 addition products
   (d) No reaction

2. When ethylene bromide is treated with Zn, we get
   (a) Alkane (b) Alkene
   (c) Alkyne (d) All

3. Ethene when treated with Br_2 in the presence of CCl_4 which compound is formed
   (a) 1, 2-dibromoethane
   (b) 1-bromo-2-chloroethane
   (c) Both (a) and (b)
   (d) 1, 1, 1-tribromoethane

4. In a reaction

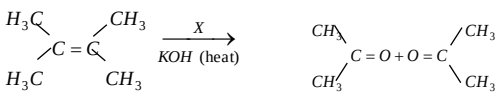
$$CH_2 = CH_2 \xrightarrow[acid]{\text{Hypochlorous}} M \xrightarrow{R} \begin{array}{l} CH_2 - OH \\ | \\ CH_2 - OH \end{array}$$

   Where *M* = molecule; *R* = reagent *M* and *R* are
   (a) CH_3CH_2Cl and *NaOH*
   (b) $CH_2Cl - CH_2OH$ and q. $NaHCO_3$
   (c) CH_3CH_2OH and *HCl*
   (d) $CH_2 = CH_2$ and heat

5. Alkenes usually show which type of reaction
   (a) Addition
   (b) Substitution
   (c) Elimination
   (d) Superposition

6. The propene reacts with *HBr* to form
   (a) Ethane
   (b) Hexane
   (c) 1-bromo-propane
   (d) 2-bromo propane

7. Ethylene may be obtained by dehydration of which of the following with concentrated H_2SO_4 at $160 - 170^oC$
   (a)C_2H_5OH
   (b) CH_3OH
   (c) $CH_3CH_2CH_2OH$
   (d) $(CH_3)_2CHCH_2OH$

8. 

   X in the above reaction is
   (a) HNO_3 (b) O_2

(c) O_3 (d) $KMnO_4$

9. The disappearance of the characteristic purple colour of $KMnO_4$ in its reaction with an alkene is the test for unsaturation. It is known as

(a) Markownikoff's test

(b) Baeyer's test

(c) Wurtz's test

(d) Grignard test

10. A gas formed by the action of alcoholic *KOH* on ethyl iodide, decolourises alkaline $KMnO_4$. The gas is

(a) C_2H_6 (b) CH_4

(c) C_2H_2 (d) C_2H_4

11. $CH_3 - CH_2 - Cl \xrightarrow{alc.KOH} A$, the product is

(a) CH_3CH_2OK (b) CH_3CHO

(c) $CH_3CH_2OCH_2CH_3$ (d) $CH_2 = CH_2$

12. The final product formed when ethyl bromide is treated with excess of alcoholic *KOH* is

(a) Ethylene

(b) Ethane

(c) Ethyne

(d) Vinyl bromide

13. Which of the following hydrocarbons cannot be obtained by Sabatier and Senderen's reaction

(a) CH_4 (b) C_2H_6

(c) C_3H_8 (d) All

14. When 3, 3-dimethyl-2-butanol is heated with H_2SO_4 the major product obtained is

(a) *cis* and *trans* isomers of 2, 3-dimethyl-2-butene

(b) 3, 3-dimethyl-1-butene

(c) 2, 3-dimethyl-2-butene

(d) 2, 3-dimethyl-1-butene

15. The intermediate during the addition of *HCl* to propene in the presence of peroxide is

(a) $CH_3C\dot{H}CH_2Cl$

(b) $CH_3\overset{+}{C}HCH_3$

(c) $CH_3CH_2C\dot{H}_2$

(d) $CH_3CH_2\overset{+}{C}H_2$

16. $CH_2 = CH_2 \xrightarrow[KOH/H_2O]{KMnO_4} X$. Product '$X$' in above reaction is

(a) Ethylene glycol (b) Glucose

(c) Ethanol (d) All of these

17. Which of the following compounds represents acrylonitrile

(a) Vinyl cyanide

(b) Cyanoethene

(c) Prop-2-ene nitrile

(d) All of them

18. When acetylene reacts with arsenic trichloride in the presence of anhydrous aluminium chloride, it produces

(a) Lewisite

(b) β-chlorovinyl dichloroarisine

(c) Nitrobenzene

(d) Both (a) and (b)

19. Ozonolysis of which one of the following will give two molecules of acetaldehyde

(a) 1-butene (b) 2-butene

(c) 1-pentene (d) 2-pentene

(e) None of these

20. In which of the following, addition of HBr does not take place against Markownikoff's rule **or** Anti-Markownikoff addition of HBr is not observed for

(a) Propene

(b) But-1-ene

(c) But-2-ene

(d) Pent-2-ene

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